Chaitanya Jadhav

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SKILLS

Programming Languages	Python, JavaScript, C#
Machine Learning	PyTorch, Scikit-Learn, NumPy, Pandas
LLMs / RAG / NLP	LangChain, FastMCP, Pydantic AI
Web & Backend	FastAPI, ASP.NET, PostgreSQL, Microsoft SQL Server
Cloud	AWS, GCP

EDUCATION

Nanyang Technological University *Aug 2021 – May 2025*
Bachelor of Computing in Computer Science GPA: 4.46/5.0

WORK EXPERIENCE

PhillipCapital Pte Ltd Jul 2025 – Present
AI Engineer

- Replaced Turi Create with a NumPy-based item-to-item collaborative filtering system for equity recommendations across SG, US, JP, and HK markets.
- Reduced churn-model training time by 50% and inference time by 72% by refactoring four scikit-learn models into per-model serialized pipelines with scheduled retraining, eliminating data leakage, and standardizing train/test preprocessing.
- Scaled internal AI chatbot adoption from 30 to 250 monthly active users by migrating to a Claude-powered MCP tool architecture on FastAPI, implementing role-based access control and multi-company data isolation, and launching authentication support for overseas offices.

Panasonic R&D Center Singapore Jan 2024 – Jul 2024
AI Research Intern

- Built an internal document Q&A chatbot using RAG, Mixtral-8x7B, BAAI/bge-en-large embeddings, Ollama, and Unstructured, with configurable chunking, retrieval, and answer-generation strategies for internal knowledge access.
- Designed a RAG benchmarking framework to compare embedding models, chunking strategies, re-rankers, and generation models on SQuAD 2.0 and internal documents, using LLM-as-a-judge evaluation to scale answer-quality assessment.

PUBLICATIONS

C. Liu, L. Qian, **C. D. Jadhav**, J. Zhao, "Joint Optimization of Secure and Energy-Efficient Retrieval-Augmented Generation for Mobile Edge Computing," *IEEE INFOCOM*, 2026. *(Accepted)*

C. D. Jadhav, C. Liu, J. Zhao, "Legal Retrieval-Augmented Generation with Structured Retrieval and Iterative Refinement," *PST 2025*.

J. Q. J. Tan, **C. D. Jadhav**, J. Komar, "The Centre That Moves: A Data-Driven Perspective on Spatial Adaptations in Men's Singles Badminton," *ISCSS 2025*.

ACADEMIC PROJECTS

Undergraduate Research Experience on Campus Program Aug 2023 – Jun 2024
Badminton Analytics with Computer Vision and ML

- Built an end-to-end badminton analytics pipeline in Python using OpenCV, OpenPose, and homography transformation to isolate courts, filter replay footage, and map player positions onto a standardized court plane for match analysis.
- Achieved 0.954 F1 on badminton grip classification by benchmarking RNN, LSTM, and Conv2D models across forehand, backhand, and overhead shot features; presented findings at the International Conference of Undergraduate Research 2024.

PERSONAL PROJECTS

Kinnect kinnect-sgsl.web.app

- Developing a Singapore Sign Language learning and translation platform with a 1,289-sign dictionary, guided lessons, and text-to-sign translation, using the OpenAI API and GCP for hosting.

Sports Event Trackers event-tracking-jkomar.pythonanywhere.com

- Built and deployed a multi-sport event-tracking web app for tennis, badminton, football, basketball, and floorball, enabling coordinate mapping, action tagging, CSV/PDF export, and keyboard-shortcut workflows for match analysis.